

Zhaoheng (Billy) Li

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Research	Systems for AI and ML; Interactive Data Analytics; Vector Databases; Graph Algorithms	
Education	Ph.D., Computer Science University of Illinois at Urbana Champaign • Advisor: Yongjoo Park	Aug 2021 - May 2026 (expected)
	B.S., Statistics and Computer Science University of Illinois at Urbana Champaign • <i>Magna Cum Laude and with Highest Distinction in Statistics and Computer Science</i> • <i>High Distinction in Mathematics</i> • Thesis: REFORM: Fast and Adaptive Solution for Subteam Replacement • Advisor: Hanghang Tong	Aug 2017 - May 2021
Professional Experience	Research Assistant University of Illinois at Urbana Champaign CreateLab	Aug 2021 - Present
	Research Intern ByteDance Inc. Systems Infrastructure Lab	May 2025 - Aug 2025
	Research Intern ByteDance Inc. Systems Infrastructure Lab	May 2024 - Oct 2024
	Software Engineer Intern Google LLC Google Cloud, Google BigQuery	May 2023 - Aug 2023
	Software Engineer Intern Google LLC Google Cloud, Systems & Storage Infrastructure	May 2022 - Aug 2022
	Research Assistant University of Illinois at Urbana Champaign IDEA Lab	Aug 2020 - May 2021
	Software Engineer Intern Google LLC Google Ads, Google Local Services	May 2020 - Aug 2020
	Software Engineer Intern Google Canada Corporation Google Ads, Google Ads Experiments	May 2019 - Aug 2019

Awards	Best Demo Award , SIGMOD 2025 <i>With Supawit Chockchowwat, Hanxi Fang, and Yongjoo Park</i>	June 2025
	Travel Grant , PyCon 2025	March 2025
	Travel Award , ICDE 2023	March 2023
	Finalist , C1Games Terminal Game AI competition	September 2019
	Best Data Visualization Award , Illini Datathon	February 2019
	Top in China , Cambridge CIE AS Sociology exam	October 2016
	National Finalist , High School Mathematical Contest in Modeling (HIMCM)	February 2015

Publications

1. Raunak Shah, **Zhaoheng Li**, Yongjoo Park
[QStore: Quantization-Aware Compressed Model Storage](#)
VLDB 2026 (research): 52nd International Conference on Very Large Databases, Boston, USA, 2026
2. **Zhaoheng Li**, Silu Huang, Wei Ding, Yongjoo Park, Jianjun Chen
[SIEVE: Effective Filtered Vector Search with Collection of Indexes](#)
VLDB 2025 (research): 51st International Conference on Very Large Databases, London, United Kingdom, 2025
3. **Zhaoheng Li**, Supawit Chockchowwat, Ribhav Sahu, Arret Sheth, Yongjoo Park
[Kishu: Time-traveling for Computational Notebooks](#)
VLDB 2025 (research): 51st International Conference on Very Large Databases, London, United Kingdom, 2025
4. **Zhaoheng Li**, Supawit Chockchowwat, Hanxi Fang, Yongjoo Park
[Demo of Kishu: Time-traveling for Computational Notebooks](#)
(**SIGMOD 2025 Best Demo Award**)
SIGMOD 2025 (demo track): ACM Special Interest Group on Management of Data, Berlin, Germany, 2025
5. Mingyuan Wu, Meitang Li, Jingcheng Yang, Jize Jiang, Kaizhuo Yan, **Zhaoheng Li**, Hanchao Yu, Minjia Zhang, Klara Nahrstedt
[A-ha Moment Revisited: Are VLMs Truly Capable of Self Verification in Inference-time Scaling?](#)
NeurIPS 2025 (short paper): Multimodal Algorithmic Reasoning workshop, Thirty-Ninth Annual Conference on Neural Information Processing Systems, San Diego, USA, 2025
6. Mingyuan Wu, Jize Jiang, Haozhen Zheng, Meitang Li, **Zhaoheng Li**, Beitong Tian, Bo Chen, Yongjoo Park, Minjia Zhang, Chengxiang Zhai, Klara Nahrstedt
[Cache-of-Thought: Master-Apprentice Framework for Cost-Effective Vision Language Model Reasoning](#)
EMNLP 2025 (main conference): 30th Conference on Empirical Methods in Natural Language Processing, Suzhou, China, 2025
7. **Zhaoheng Li**, Pranav Gor, Rahul Prabhu, Hui Yu, Yuzhou Mao, Yongjoo Park

ElasticNotebook: Enabling Live Migration for Computational Notebooks

VLDB 2024 (research): 50th International Conference on Very Large Databases, Guangzhou, China 2024

8. **Zhaoheng Li**, Supawit Chockchowwat, Hanxi Fang, Ribhav Sahu, Sumay Thakurdesai, Kantanat Pridaphatrakun, Yongjoo Park

Demo of ElasticNotebook: Migrating Live Computational Notebook States

SIGMOD 2024 (demo track): ACM Special Interest Group on Management of Data, Santiago, Chile, 2024

9. Supawit Chockchowwat, **Zhaoheng Li**, Yongjoo Park

Transactional Python for Durable Machine Learning: Vision, Challenges, and Feasibility

SIGMOD DEEM 2023 (short paper): 7th Workshop on Data Management for End-to-End Machine Learning, Seattle, USA, 2023

10. **Zhaoheng Li**, Xinyu Pi, Yongjoo Park

S/C: Speeding up Data Materialization with Bounded Memory

ICDE 2023 (research): 39th International Conference on Data Engineering, Anaheim, USA, 2023

11. **Zhaoheng Li**, Xinyu Pi, Mingyuan Wu, Hanghang Tong

REFORM: Fast and Adaptive Solution for Subteam Replacement

IEEE BigData 2021 (research): IEEE International Conference on Big Data, 2021

12. **Zhaoheng Li**, Xinyu Pi, Mingyuan Wu

Subteam Replacement: Problem Definition and Fast Solution

SIGMOD SRC 2021 (extended abstract): ACM Special Interest Group on Management of Data Student Research Competition, Xi'an, China, 2021

Teaching	ML and Data Systems (CS 598) Teaching Assistant, University of Illinois at Urbana-Champaign	Spring 2024
	Game Development (CS 498) Teaching Assistant, University of Illinois at Urbana-Champaign	Spring 2022
	Introduction to Computer Science (CS 125) Course Assistant, University of Illinois at Urbana-Champaign	Spring 2018
Invited Talks	JupyterCon 2025 , San Diego, USA	November 2025
	VLDB 2025 , London, United Kingdom	September 2025
	SIGMOD 2025 , Berlin, Germany	June 2025
	PyCon 2025 , Pittsburgh, USA	May 2025
	UIUC NCSA , Urbana, USA	April 2025
	VLDB 2024 , Guangzhou, China	August 2024
	Bytedance Inc. , San Jose, USA	August 2024

SIGMOD 2024, Santiago, Chile

June 2024

ICDE 2023, Anaheim, USA

April 2023

IEEE BigData 2021 (Virtual)

December 2021

SIGMOD SRC 2021, Xi'An, China (Virtual)

June 2021